Core-energy concentration in rotationally superposed apodized catastrophe fields

Auto BeamScientist Project

Abstract

Rotational superposition changes both the symmetry and the energy distribution of a structured optical field. We study this redistribution for Gaussian-apodized Pearcey and butterfly fields in scalar free-space propagation. Decomposing a field into inversion-even and inversion-odd components shows that coherent addition to its 180° rotated copy selects the even component. The resulting change in power inside a circular observation region depends on the concentration of the two components, rather than on a new propagation law. We combine this description with equal-budget searches over a common bounded phase family. After fixing the selected parameters, the Pearcey and butterfly compositions reach median core-energy fractions of 5.965% and 5.901%, respectively, compared with 3.970% for the strongest Gaussian control with the same phase bounds and composition options. Mesh refinement, an expanded window, denser axial sampling, and direct Fresnel quadrature test the numerical observations. An ideal Gaussian thin-lens reference is reported separately because it exceeds the imposed quadratic-phase range. These results support a conditional concentration advantage within the stated field and phase classes; the advantage does not establish superiority over unrestricted focusing optics.

1 Introduction

The usefulness of a structured optical field depends on where its energy is delivered. A complex intensity pattern can contain a bright local feature while placing little input power inside a prescribed region. Core-energy concentration therefore provides a complementary description to peak intensity, the position of the brightest pixel, or the visual appearance of a caustic. When coherent copies are combined, normalization and interference must be considered together to determine whether the energy delivered to that region actually increases.

Canonical diffraction integrals provide a mathematical description of optical catastrophe fields. Pearcey beams connect the cusp diffraction integral to transverse optical fields and have been investigated for autofocusing and reconstruction after obstruction [3]. Higher-order integrals provide swallowtail and butterfly fields through particular sections of their control-parameter spaces [9]. The finite aperture, apodization, and chosen section determine the field used in a numerical calculation. Consequently, a property of one section does not establish a universal dependence on catastrophe order.

We consider the coherent superposition of a field with a copy rotated through 180°. This operation has a direct symmetry interpretation: the sum retains the inversion-even component and removes the inversion-odd component. For an inversion-even Gaussian field, the normalized sum reproduces the original field. For a mixed-parity field, it changes the input distribution before propagation. This distinction makes rotational pairing a useful setting in which to separate an established linear operation from a field-dependent concentration result.

Two comparisons are needed. The first asks whether a particular structured field delivers more core energy after rotational pairing than before it. The second asks whether the paired field improves on optimized classical controls under stated constraints. These questions have different denominators. A large paired-versus-single ratio can occur for a field with low absolute concentration. Conversely, a Gaussian with an identical rotated copy has a ratio of unity even

when its absolute concentration is high. Optimizing the ratio alone would therefore provide an inadequate classical benchmark.

Our study begins with two positive observations selected by Auto BeamScientist from an autonomous numerical exploration. We then freeze a follow-up objective and compare absolute concentration using a shared optimizer, common additional-phase bounds, equal trial counts, and Gaussian controls with both composition options. Numerical checks are performed with the selected parameters held fixed. The resulting study concerns energy redistribution in specified finite-aperture fields. It also identifies the distinction between performance under bounded additional phase and conventional thin-lens focusing, which is essential for interpreting the measured advantage.

2 Related work

2.1 Catastrophe fields and Pearcey-beam engineering

Pearcey-beam research connects a canonical cusp integral to optical propagation, including autofocusing and recovery after obstruction [3]. Higher-order catastrophe fields extend this construction through selected sections of the swallowtail and butterfly control spaces [9]. These studies provide the field definitions underlying the present apodized seeds. Our butterfly source uses one particular two-control section with the remaining controls fixed at zero, so its numerical comparison concerns that specified field rather than every butterfly configuration.

Symmetry and superposition are established tools for engineering Pearcey-derived fields. Symmetric Pearcey Gaussian beams combine a Gaussian envelope with two multiplying Pearcey integrals and exhibit adjustable focusing and transverse structure [7]. Masked Pearcey beam arrays combine multiple Pearcey fields with aperture masking to modify autofocus behavior [6]. These constructions are close in their use of structured seeds and field composition. The composition studied here is instead the additive pair $U(x, y) + U(-x, -y)$, followed by equal-power normalization. We evaluate its median energy inside a fixed circular region over a common propagation interval, which distinguishes this objective from peak focusing intensity or the focal position of an array.

Propagation-medium engineering offers a complementary control route. Analytical propagation of Pearcey fields in uniaxial crystals relates autofocus and inversion positions to the extraordinary refractive index [8]. The present calculation fixes homogeneous free space and varies the boundary field and its additional phase. The distinction matters because a medium-dependent trajectory or focus is a different physical problem from concentration under a fixed propagation operator.

2.2 Rotational interference and spatial-parity selection

Interference between a field and its rotated copy has an established interpretation in spatial-mode optics. Two-dimensional parity sorting has been demonstrated with an out-of-plane Sagnac interferometer whose counter-propagating fields acquire a relative 180° transverse rotation [1]. The two output ports separate inversion-even and inversion-odd spatial components. The projection $U \mapsto U + \mathcal{R}U$ used below is the even-channel operation in this description.

Our analysis uses this known projection to interpret core-energy redistribution in the chosen catastrophe fields. The numerical comparison normalizes the selected field to a prescribed output power and evaluates its subsequent propagation. A physical parity sorter also distributes power between output ports, so normalized field concentration and complete optical-system throughput remain distinct quantities. This distinction motivates the explicit normalization and the Gaussian control with the same coherent-composition operation.

2.3 Phase reshaping and the present comparison

Continuous phase-only reshaping can transform structured light toward a Gaussian profile and improve its beam-quality factor M^2 [5]. That work illustrates how an optical transformation can improve a chosen concentration-related property while preserving power in the modeled transformation. Here the measured quantity is the median energy in a specified on-axis core, rather than M^2 or laser brightness. The additional-phase family is also restricted to common finite coefficient ranges.

The present study brings the source definition, rotational composition, and bounded phase search into one fixed-power numerical comparison. It includes both single-field and coherently composed Gaussian references, optimized defocus and quadratic–quartic controls, and a separate ideal-lens reference. The reported margins therefore describe the selected finite-aperture field classes under the stated constraints. The parity projection supplies the physical interpretation, while absolute core energy supplies the common performance measure.

3 Fields, propagation, and rotational projection

3.1 Finite-aperture field definitions

Let $\mathbf{r} = (x, y)$ be the transverse coordinate and let $r^2 = x^2 + y^2$. We use the canonical integrals

$$C_n(u, v) = \int_{-\infty}^{\infty} \exp\big[i\left(t^n + vt^2 + ut\right)\big]\, dt, \qquad n \in \{4, 6\}. \tag{1}$$

The case $n = 4$ is the Pearcey integral. The case $n = 6$ is a two-control section of the butterfly integral, with the coefficients of t^3 and t^4 fixed at zero [9]. These are mathematical field definitions; their behavior is evaluated for the apodized fields below.

The seed fields are

$$U_n(\mathbf{r}) = A_n\, \mathbf{1}_{r\le a} \exp(-r^2/w^2)\, C_n(x/s, y/s), \qquad U_{\mathrm{G}}(\mathbf{r}) = A_{\mathrm{G}}\, \mathbf{1}_{r\le a} \exp(-r^2/w^2). \tag{2}$$

Here a is the aperture radius, w is the Gaussian-envelope radius, and s is the canonical-coordinate scale. The constants A_n and A_{G} give the same sampled input power P:

$$\Delta x^2 \sum_{j,k} |U(x_j, y_k)|^2 = P. \tag{3}$$

For the catastrophe fields, the envelope width is w; an additional field named apodization in the exploration schema does not change this envelope in the implemented generator. We use the actual generator definition in Eq. (2).

Each source is modified by the same additional-phase family,

$$\Phi(x, y) = q_2\rho^2 + q_3(u_0^3 + v_0^3) + q_4\rho^4 + \eta \sin(2\rho^2), \quad u_0 = x/w, \quad v_0 = y/w, \quad \rho^2 = u_0^2 + v_0^2. \tag{4}$$

All coefficients are dimensionless. The coefficient ranges are given in Sec. 4. These constraints apply to the additional phase, not to the full intrinsic phase of a canonical seed. Different seeds consequently represent different allowed complex-field distributions.

3.2 Scalar propagation and the measured core energy

The sampled angular-spectrum operator is

$$\mathcal{L}_z U = \mathcal{F}^{-1}\{\mathcal{F}[U] H_z\}, \tag{5}$$

$$H_z(f_x, f_y) = B_z(f_x, f_y) \exp\left[i2\pi z \sqrt{\lambda^{-2} - f_x^2 - f_y^2}\right], \tag{6}$$

$$B_z = \mathbf{1}_{f_x^2 + f_y^2 \le \min(f_c(z)^2, \lambda^{-2})}, \qquad f_c(z) = \frac{\lambda^{-1}}{\sqrt{1 + (2z/L)^2}}. \tag{7}$$

Here L is the transverse window width, λ is the wavelength, and f_x, f_y are spatial frequencies in cycles per metre. Equation (7) is a conservative radial truncation used by this implementation. It is distinct from the complete band-limited angular-spectrum construction [2].

The independent calculation uses separable real-space Fresnel quadrature,

$$U_z^{\mathrm{F}}(x,y) = \frac{e^{ikz}}{i\lambda z} \iint U(x',y') \exp\left[\frac{ik}{2z}\big((x-x')^2 + (y-y')^2\big)\right] dx'\, dy', \quad k = 2\pi/\lambda. \tag{8}$$

This checks the paraxial regime through a different numerical operator. Agreement in the present low-angle setting does not extend the calculation to vector or near-field propagation.

For a fixed circular region $D_c = \{r \le r_c\}$, the plane-resolved core fraction and the optimization objective are

$$c_U(z) = \frac{1}{P} \iint_{D_c} |\mathcal{L}_z U|^2 dx\, dy, \qquad \mathcal{C}[U] = \mathrm{median}_{z\in\mathcal{Z}}\, c_U(z). \tag{9}$$

The observation region stays on the optical axis and is identical for every method. It does not follow a bright spot. The denominator is input power, and the median is evaluated on the recorded finite axial grid $\mathcal{Z}$.

3.3 Parity interpretation

Define the rotation operator $\mathcal{R}U(x,y) = U(-x,-y)$ and the components

$$U_+ = \frac{U + \mathcal{R}U}{2}, \qquad U_- = \frac{U - \mathcal{R}U}{2}, \qquad p_\pm = \frac{\|U_\pm\|^2}{P}. \tag{10}$$

The two components are orthogonal over an inversion-symmetric domain, and $p_+ + p_- = 1$. When $p_+ > 0$, the equal-power coherent composition is

$$U_{\mathrm{coh}} = \frac{U + \mathcal{R}U}{\sqrt{2(1 + \mathrm{Re}\,\gamma)}} = \frac{U_+}{\sqrt{p_+}}, \qquad \gamma = \frac{\langle U, \mathcal{R}U\rangle}{P}. \tag{11}$$

This definition compares synthesized output fields with equal power. A passive interferometer would also have a rejected output port and insertion losses; Eq. (11) does not measure source-to-target conversion efficiency.

In free space, rotation commutes with propagation. The circular aperture and observation region preserve the same symmetry. The integral of the interference term between an even and an odd component therefore vanishes over D_c. If $c_\pm(z)$ denote the core fractions of the individually normalized components, then

$$c_U(z) = p_+ c_+(z) + p_- c_-(z), \qquad c_{U_{\mathrm{coh}}}(z) - c_U(z) = p_-\big[c_+(z) - c_-(z)\big]. \tag{12}$$

Pairing increases core energy when the even component is more concentrated than the odd component at that plane. The sign depends on the field and the chosen region. Equation (12) applies plane by plane; medians of separate curves must be evaluated before comparing the reported objectives.

The equal-power incoherent mixture has intensity

$$I_{\mathrm{inc}}(\mathbf{r}, z) = \tfrac{1}{2}\left(|\mathcal{L}_z U(\mathbf{r})|^2 + |\mathcal{L}_z \mathcal{R}U(\mathbf{r})|^2\right). \tag{13}$$

Its integrated circular-core energy equals that of the single field in this symmetric geometry. It is useful as an intensity-level control but supplies the same integrated reference for Eq. (9). An exactly inversion-odd input has $p_+ = 0$ and zero coherent sum. Renormalizing this sum to finite power is undefined, so such cases are recorded as cancellation outcomes rather than concentration successes.

4 Numerical study

4.1 Exploration and follow-up selection

The initial autonomous campaign recorded 665 hypotheses and 13,433 physics-evaluation invocations. It completed 129 cycles with scientific critiques and saved 600 numerical array files. The campaign contains repeated questions, correlated candidates, known identities, and failed runs; these counts are computational provenance rather than independent statistical samples. The two observations used here were selected after inspecting that campaign. They are not a prospectively selected discovery or an independent estimate of the campaign's success rate.

In the selected cycle, the original numerical fields used free-space propagation. Its natural-language plan referred to a thin lens, but the executed environment omitted the lens phase. We therefore interpret the stored results as free-space core-energy measurements. The follow-up explicitly freezes that geometry and does not use the stored brightest-plane coordinate to claim an autofocus position.

The common numerical conditions are $\lambda = 633$ nm, $P = 1$ W, $a = 0.9$ mm, $w = 0.30$ mm, $s = 0.15$ mm, and $r_c = 35$ μm. The base window is $L = 2.4$ mm. The axial grid contains 21 uniformly spaced planes over $z = 49$–71 mm. Catastrophe integrals are evaluated using a rotated contour $t = \tau \exp[i\pi/(2n)]$ and trapezoidal quadrature; the source generator is checked by increasing the contour sample count from 192 to 384.

4.2 Equal-budget phase search and classical controls

We search each source with and without coherent rotational composition. Gaussian controls use the same two composition options, so an advantage due simply to adding a second coherent arm is not attributed to catastrophe order. Every four-parameter arm receives 64 proposal evaluations from the same bounded numerical optimizer and seed 103. The search uses nested initial samples followed by differential mutation, jitter, and seeded restarts. Selection is performed at $N = 384$, and all parameters are then fixed for verification.

The shared additional-phase bounds are

$$q_2 \in [-1.2, 1.2], \quad q_3 \in [-0.4, 0.4], \quad q_4 \in [-0.7, 0.7], \quad \eta \in [-1, 1]. \tag{14}$$

Gaussian defocus and quadratic-plus-quartic controls are also optimized with 64 trials each. The objective is the absolute $\mathcal{C}$ in Eq. (9), rather than a normalized paired-versus-single ratio. Two catastrophe seeds and two composition options consume more aggregate search work than one Gaussian arm; the comparison equalizes the per-arm budget and does not measure an LLM-specific optimization benefit. The initial fields also differ in amplitude and intrinsic phase, so the result compares specified field classes rather than equal-cost physical field generators.

An ideal Gaussian lens with $f = 60$ mm is included as an additional reference. Its phase has $q_2 = -\pi w^2/(\lambda f) \simeq -7.44$, outside Eq. (14). This reference tests the boundary of a bounded-phase claim. It is not counted as an admissible sample in the constrained searches.

4.3 Numerical and perturbation checks

Verification uses $N = 384, 576, 768$, and 1152 at the original window width. Window expansion uses $L = 3.6$ mm and $N = 1728$, preserving the pixel pitch of the $N = 1152$ calculation. Axial refinement uses 41 planes. Direct Fresnel quadrature is evaluated on the full $N = 768$ grid and the common 21 planes. These tests use fixed selected coefficients rather than refitting each numerical configuration.

The physical checks include eight-level input-phase quantization, an independent 0.02-rad RMS Gaussian phase perturbation on the pre-composition field with seeds 103, 107, and 113, and a held-out wavelength of 640 nm. The complex boundary field and the selected phase coefficients

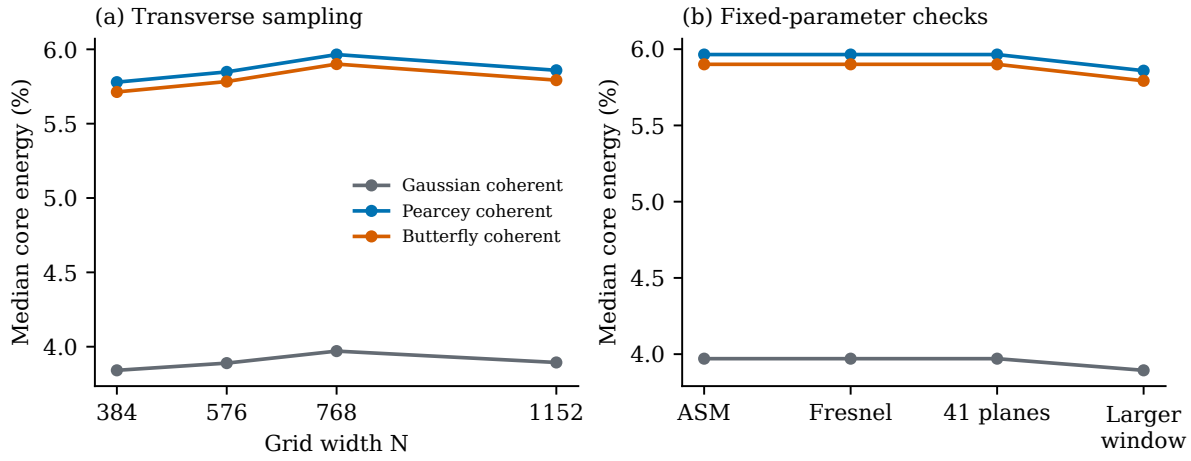


Figure 1: Fixed-parameter verification of absolute core concentration. Left: the median core fraction as the transverse grid is refined. Right: the same frozen coefficients evaluated with angular-spectrum propagation, real-space Fresnel quadrature, a denser axial grid, and a larger transverse window. Values are deterministic numerical measurements, without error bars or independent-replicate interpretation.

are held fixed for the wavelength check. This convention tests a changed propagation wavelength and does not model a fabricated plate's chromatic optical-path response. Quantization is applied before coherent composition; the summed output field can consequently have phase values between the quantization levels.

Every numerical record includes input-power checks, a minimum retained spectral-power threshold of 0.995, and a maximum boundary-energy fraction of 0.01. Numerical agreement uses a predeclared 3% relative tolerance for core fractions, and a positive follow-up effect requires a relative paired-versus-single increase of at least 5%. A separate 3% margin tests superiority over the strongest bounded Gaussian control. These tolerances are deterministic numerical criteria and do not represent statistical confidence intervals.

5 Results and physical interpretation

5.1 Stored observations and the effect of core sampling

The selected campaign records gave single-to-coherent median core fractions of $0.0301524 \rightarrow 0.0328636$ for Pearcey and $0.0309919 \rightarrow 0.0332519$ for butterfly. The corresponding relative increases were 8.99% and 7.29%. These values used $N = 192$, for which the 35-μm core contains 21 grid samples. Their positive sign motivated the follow-up, but their absolute values were sensitive to the discretized core area.

The first follow-up confirmed the positive sign using a different propagation operator and a larger computational window, while the $N = 256$ to 384 core-fraction change exceeded the 3% criterion. This result motivated the finer fixed-parameter verification, rather than treating a source-integral quadrature check as proof of image-grid convergence. The final mesh comparison and the independent-operator agreement are shown in Fig. 1.

5.2 Absolute concentration under the common phase bounds

Table 1 reports the selected methods on the $N = 768$ verification grid. The optimized Pearcey and butterfly coherent compositions reach 5.965% and 5.901%, respectively. The strongest admissible Gaussian control across both composition options reaches 3.970%. Relative to that

Table 1: Absolute median core energy on the $N = 768$ verification grid. Each bounded four-parameter arm has 64 search trials at $N = 384$. The lens reference is outside the quadratic-phase constraint.

Field and composition	Core energy (%)	Gain over bounded Gaussian (%)
Gaussian single	3.9695	-0.02
Gaussian coherent	3.9704	+0.00
Pearcey single	5.5154	+38.91
Pearcey coherent	5.9648	+50.23
Butterfly single	5.5302	+39.29
Butterfly coherent	5.9008	+48.62
Optimized Gaussian defocus	3.7658	-5.15
Optimized Gaussian quadratic–quartic	3.8937	-1.93
Gaussian, ideal 60-mm lens (unbounded reference)	64.9869	—

Table 2: Frozen selected additional-phase coefficients. All four coefficients use the same dimensionless convention in Eq. (4).

Field and composition	q_2	q_3	q_4	η
Gaussian single	-1.200000	-0.023394	-0.545840	-0.032055
Gaussian coherent	-1.200000	-0.023394	-0.545840	-0.032055
Pearcey single	-1.200000	-0.033557	-0.698181	-0.038097
Pearcey coherent	-1.199039	-0.019267	-0.700000	-0.035694
Butterfly single	-1.200000	-0.033557	-0.698181	-0.038097
Butterfly coherent	-1.199039	-0.019267	-0.700000	-0.035694

control, the gains are 50.23% and 48.62%. These comparisons use absolute core energy and avoid the unity-ratio degeneracy of a centrosymmetric Gaussian and its rotated copy.

The single-field structured controls also concentrate more energy than the bounded Gaussian controls. Rotational composition provides an additional change, whose magnitude must be assessed relative to each source's independently optimized single-field result. Table 2 gives the selected coefficients; the same coefficients are used in every numerical verification. Figure 2 shows the saved longitudinal marginal profiles for the final structured-field comparisons, using common coordinates and a shared intensity scale within each family.

The ideal-lens Gaussian reference reaches 64.99% under the same core and observation window. It exceeds the structured-field values and uses a quadratic phase beyond the search bounds. The positive comparison in Table 1 is consequently a result for the specified bounded additional-phase classes, rather than an improvement over conventional unrestricted focusing. This distinction also prevents a small Gaussian paired-versus-single ratio from being misinterpreted as a weak absolute focusing baseline.

5.3 Perturbations and interpretation of the advantage

Figure 3 reports the selected fields under three fixed noise realizations and the held-out wavelength. These are separate deterministic conditions rather than an empirical distribution of fabrication errors. The positive comparison with the bounded Gaussian control persists in these conditions. The eight-level phase trials retain less than the required 99.5% spectral power for every arm and are excluded from this figure and from the confirmed comparisons. They remain diagnostic records requiring a separate bandwidth assessment.

The parity description explains why the normalized sum can differ from a single non-centrosymmetric source, while a centrosymmetric Gaussian with radial phase remains unchanged.

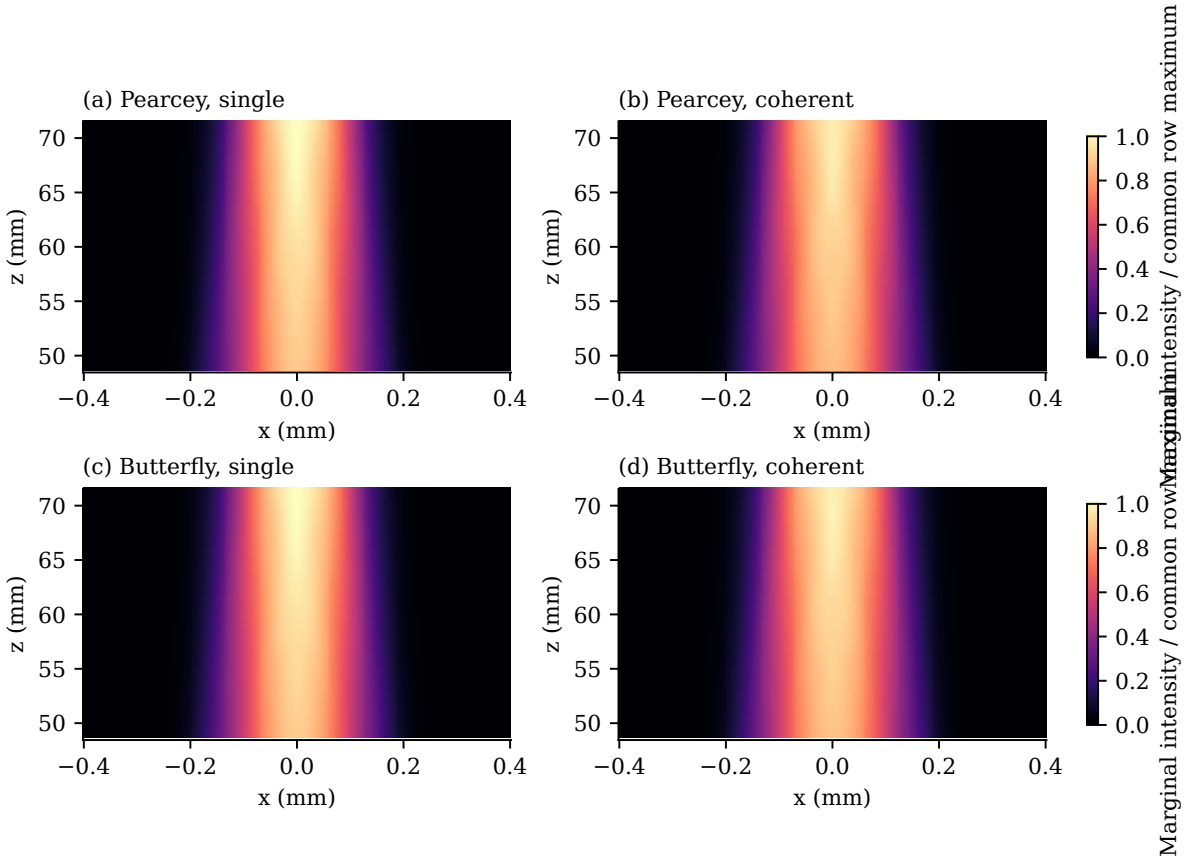


Figure 2: Longitudinal marginal intensity for the optimized single-field and coherently composed catastrophe sources. The displayed quantity is $M(x,z) = \int_{|y|<0.6\,\mathrm{mm}} I(x,y,z)dy$, taken from saved $N = 768$ arrays on the common axial grid. Each row uses a common normalization for the two composition states. The profiles describe transverse redistribution; they are not on-axis trajectories or proof of an interior autofocus.

Equation (12) specifies the condition for increased circular-core energy without introducing a new beam family. The concentration measurements alone do not establish catastrophe-order specificity: an appropriate generic mixed-parity field can undergo the same projection. Nor do the small differences between single-field and incoherent circular-core integrals provide an independent test of a coherent mechanism, because those integrals are equal by symmetry in the ideal geometry.

The added numerical checks strengthen the evidence for the particular concentration comparisons. They leave separate physical questions about field-synthesis loss, chromatic implementation, arbitrary phase ranges, and alternative source-amplitude classes. In particular, equal synthesized output power does not imply equal laser-to-target efficiency. The advantage measured here is a property of the normalized boundary fields and the defined observation region.

6 Conclusions

Coherent 180° rotational superposition selects the inversion-even component of an optical field. Its effect on circular-core energy is governed by the relative concentration of the even and odd components. For the specified apodized Pearcey and butterfly sources, equal-budget bounded phase searches produce positive concentration results against Gaussian controls that include the same composition operation. Fixed-parameter mesh, window, axial, and independent-propagator

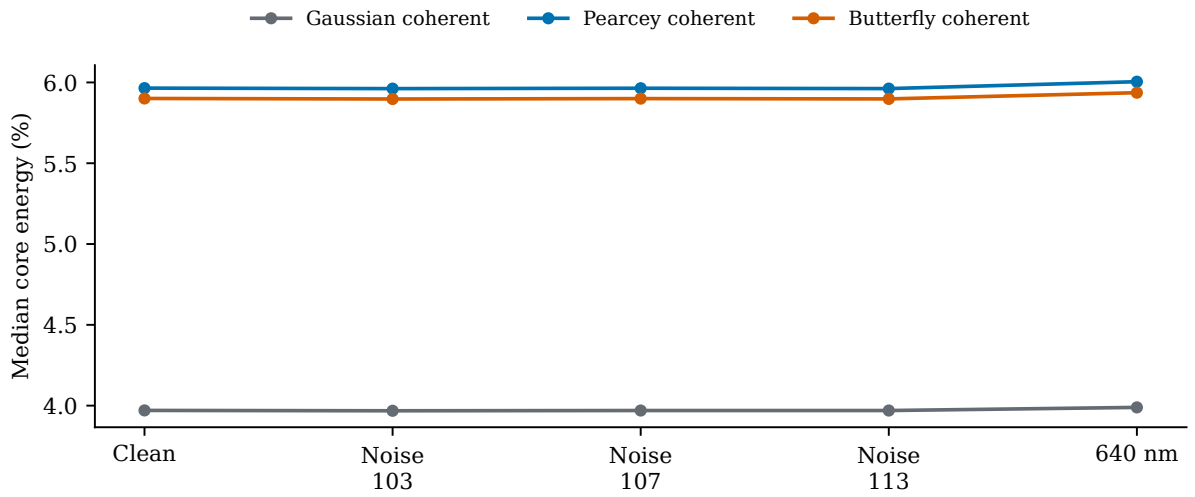


Figure 3: Core concentration for the strongest bounded Gaussian control and the two coherent catastrophe compositions under the frozen perturbation protocol. Noise labels identify the three stored seeds at 0.02-rad RMS. The wavelength check holds the input complex field fixed. Eight-level phase trials fail the spectral-retention criterion and are excluded. The plot does not imply a fabrication-tolerance distribution.

checks separate these results from the coarse-grid observations that motivated them. An ideal Gaussian lens remains stronger outside the imposed phase bounds. The supported outcome is therefore a conditional concentration result for explicit finite-aperture field classes, with a symmetry-based explanation and reproducible numerical evidence.

Data and code

The saved array profiles, phase-search trials, numerical protocols, and plotting source accompany this manuscript. Auto BeamScientist adapts Sakana AI Scientist-v2 [4] for research ideation and scientific criticism.

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